

---=Operators:

1)

```
int a = 5;
```

```
int b = 2;
```

```
System.out.println(a + b + "Hello" + a + b);
```

- A. 7Hello52
- B. 7Hello7
- C. 7Hello5 + 2
- D. Hello57

 **Answer:** A. 7Hello52

 **Explanation:**

- First, $a + b \rightarrow 5 + 2 = 7$ (numeric addition)
- Then `"7" + "Hello" = "7Hello"` (String concatenation)
- Then `"7Hello" + a \rightarrow "7Hello5"`
- Then `"7Hello5" + b \rightarrow "7Hello52"`

2)

```
System.out.println(10 + 20 + "30" + 40 + 50);
```

- A. 30 30 90
- B. 30304050
- C. 303050
- D. 303040

 **Answer:** B. 30304050

3) `int x = 5;`

```
System.out.println(x++ + ++x);
```

- A. 10
- B. 11
- C. 12
- D. 13

 **Answer:** B. 12

 **Explanation:**

- `x++` returns 5, then `x` becomes 6

- ++x increments x to 7 and returns 7
- So, $5 + 7 = 12$

4)

```
int a = 10;
```

```
int b = 20;
```

```
System.out.println(a++ > 10 && ++b > 20);
```

```
System.out.println(a + " " + b);
```

A. false 11 21

B. true 11 21

C. false 11 20

D. true 10 21

 **Answer:** C. false 11 20

 **Explanation:**

- $a++ > 10 \rightarrow 10 > 10 \rightarrow \text{false}$, but a becomes 11
- && short-circuits, ++b is **not evaluated**, so b = 20

5)

Which of the following is not a valid Java operator?

A. instanceof

B. >>>

C. ^^

D. <<

 **Answer:** C. ^^

 **Explanation:**

- ^^ is **not** a valid operator.
- ^ is bitwise XOR. There is no ^^ in Java.

6)

```
int a = 4;
```

```
System.out.println(a << 1);
```

A. 8

B. 6

C. 10

D. 4

■ **Answer:** A. 8

■ **Explanation:**

- << is **left shift**: shifts bits to the left
- 4 in binary: 0100 → left shift 1 → 1000 = 8

7)

```
boolean a = true;
```

```
boolean b = false;
```

```
System.out.println(a & b);
```

```
System.out.println(a && b);
```

- A. false false
- B. false true
- C. true false
- D. true true

■ **Answer:** A. false false

■ **Explanation:**

- & and && both return false for true and false
- & checks both sides; && short-circuits after left side if false

8)

What will be the result of the following code?

```
int a = 1;
```

```
int b = a++ + a++ + a;
```

```
System.out.println(b);
```

- A. 5
- B. 4
- C. 6
- D. 7

■ **Answer:** C. 6

■ **Explanation:**

- a++ returns 1 (then a=2)
- a++ returns 2 (then a=3)
- final a is now 3

- So $b = 1 + 2 + 3 = 6$

9)

What is the output of the code below?

```
int a = 5;
```

```
int b = 10;
```

```
int c = 15;
```

```
System.out.println(a > b ? a > c ? a : c : b > c ? b : c);
```

- A. 5
- B. 10
- C. 15
- D. Compilation error

 **Answer:** C. 15

 **Explanation:**

Ternary nested:

- $a > b \rightarrow \text{false}$
- $\text{else: } b > c \rightarrow \text{false}$
- so, $c = 15$

10)

13. What will this print?

```
int a = 4, b = 3;
```

```
System.out.println((a | b) + (a & b));
```

- A. 7
- B. 6
- C. 8
- D. 5

 **Answer:** A. 7

 **Explanation:**

- $a | b = 7$
- $a \& b = 0$
- $7 + 0 = 7$

11)

Choose the correct output:

```
int a = 2;
```

```
int b = 3;
```

```
System.out.println(b++ * ++a);
```

- A. 6
- B. 9
- C. 8
- D. 10

 **Answer:** C. 9

12)

What is the result of the following expression?

```
System.out.println(10 * 20 + "30" + 40 * 2);
```

- A. 203040
- B. 2003040
- C. 203080
- D. 2003080

 **Answer:** D. 2003080

 **Explanation:**

- $10 * 20 = 200$
- $40 * 2 = 80$
- So $200 + "30" \rightarrow "20030"$
- Then $"20030" + 80 \rightarrow "2003080"$

13)

What is the result of this loop?

```
int i = 0;
```

```
while (i++ < 5) {
```

```
    i++;
```

```
}
```

```
System.out.println(i);
```

- A. 5
- B. 6

C. 7

D. Infinite loop

■ **Answer:** C. 7

■ **Explanation:**

Loop iterations:

- $i=0 \rightarrow i++ < 5 \rightarrow \text{true} \rightarrow i=1 \rightarrow i++ = 2$
- $i=3 \rightarrow i++ < 5 \rightarrow \text{true} \rightarrow i=4 \rightarrow i++ = 5$
- $i=6 \rightarrow i++ < 5 \rightarrow \text{false}$
Final $i = 7$

14)

What happens if you divide an integer by 0 in Java?

`int x = 10 / 0;`

A. 0

B. Infinity

C. `ArithmeticException`

D. NaN

■ **Answer:** C. `ArithmeticException`

■ **Explanation:**

- Division by zero for **integers** throws **`ArithmeticException`** in Java.

15)

What is the output of the following?

`System.out.println(10 == 10.0);`

A. true

B. false

C. Compilation error

D. Runtime error

■ **Answer:** A. true

■ **Explanation:**

- 10 is int, 10.0 is double
- Implicit type promotion $\rightarrow$ both become double
- $10.0 == 10.0 \rightarrow \text{true}$

16)

Which is the correct evaluation order of this expression?

```
System.out.println(2 + 3 * 4 / 2 - 5);
```

- A 5
- B 3
- C. Left to right strictly
- D. Error in precedence

■ **Answer:** B.

$3 * 4 = 12 \rightarrow 12 / 2 = 6 \rightarrow 2 + 6 = 8 \rightarrow 8 - 5 = 3$

17) What is printed here?

```
int x = 10;
```

```
int y = 10;
```

```
System.out.println(x++ == ++y);
```

- A. true
- B. false
- C. Error
- D. Depends on compiler

■ **Answer:** B. false

■ **Explanation:**

- x++ returns 10, x becomes 11
- ++y becomes 11
- So $10 == 11 \rightarrow \text{false}$

18)

What will be the output?

```
int a = 10;
```

```
if (a++ > 10)
```

```
    System.out.println("Greater");
```

```
else if (a == 11)
```

```
    System.out.println("Equal");
```

```
else
```

```
    System.out.println("Lesser");
```

- A. Greater
- B. Equal
- C. Lesser
- D. Compilation error

 **Answer: B**

 `a++ > 10` is false since `a = 10`, then `a` becomes 11. Now, `a == 11` is true.

19)

3. Which of the following is valid?

- A. `if (true); else;`
- B. `if {true}`
- C. `if (false)`
- D. `if (1)`

 **Answer: C**

 Only `if (false)` is syntactically valid; Java doesn't allow `if (1)` like C/C++.

20)

Output of this code?

```
int x = 5;
```

```
if (x++ == 5 && ++x == 7)
```

```
    x += 3;
```

```
System.out.println(x);
```

- A. 6
- B. 7
- C. 8
- D. 10

 **Answer: D**

 `x++ == 5` → true (`x=6`), `++x == 7` → `x=7` → true → `x += 3` → `x=10`

Oops! Correction:

- `x++ = 5` → `x=6`
- `++x = 7` → true
- `x += 3` → `x = 10`

21)

What is printed?

```
int i = 5;
```



```
if (i > 1)
    if (i < 10)
        System.out.println("A");
    else if (i < 20)
        System.out.println("B");
else
    System.out.println("C");
```

- A. A
- B. B
- C. C
- D. Compilation Error

■ **Answer:** A

■ Only first if is true → prints "A"

22)

What is the Output?

```
int a = 2, b = 3;
if ((a > b) & (b++ > a))
    System.out.println(b);
else
    System.out.println(b);
```

- A. 3
- B. 4
- C. Error
- D. 2

■ **Answer:** B

■ & evaluates both sides even if first is false → b++ runs → b = 4

23)

Find the output:

```
int x = 5;
switch (x) {
    default:
        System.out.println("Default");
}
```

```
case 5:

    System.out.println("Five");
}
```

- A. Default
- B. Five
- C. Default Five
- D. Compilation Error

 **Answer: B**

24)

Find the output:

```
int x = 6;

switch (x) {

    default:

        System.out.println("Default");

    case 5:

        System.out.println("Five");

}
```

- A. Default
- B. Five
- C. Default Five
- D. Compilation Error

 **Answer: C**

 No break → fall-through from default to case 5

25)

What does this print?

```
int i = 1;

switch (i) {

    case 0:

        i++;

    case 1:

        i += 2;

}
```

```

    case 2:
        i += 3;
        break;
    default:
        i += 4;
}
System.out.println(i);

```

- A. 6
- B. 3
- C. 7
- D. 10

 **Answer: A**

 Case 1 → i = 3, then i = 6 (in case 2), then break → prints 6

26)

Will this compile?

```

int x = 0;
if (x = 1) {
    System.out.println("Hello");
}

```

- A. Yes, prints Hello
- B. Compilation error
- C. Runtime error
- D. Prints nothing

 **Answer: B**

 Cannot assign (=) in if() → must be a boolean condition

27)

Output?

```

int x = 10;
while (x-- > 5)
    if (x == 7)
        break;
System.out.println(x);

```

- A. 7
- B. 6
- C. 5
- D. Infinite loop

 **Answer: 7**

 x=10 → x=9 → 8 → 7 → break → print x = 7

28)

Choose correct statement:

```
switch (5) {
    case 1:
    case 2:
    default:
    case 3:
        System.out.println("Hi");
}
```

- A. Hi
- B. Compilation error
- C. Nothing
- D. Infinite loop

 **Answer: A**

 No break, but case 5 doesn't match → default triggers → prints "Hi"

30)

Output of this:

```
int x = 0;

do {
    x++;
    continue;
} while (x < 3);

System.out.println(x);
```

- A. 1
- B. 2
- C. 3
- D. Infinite loop

■ **Answer: C**

■ continue skips rest, but loop runs until $x < 3 \rightarrow x$ becomes 3

31)

Will this run?

```
while (false) {  
    System.out.println("Hi");  
}
```

- A. Yes
- B. No
- C. Compilation Error
- D. Infinite Loop

■ **Answer: A**

■ While false $\rightarrow$ loop never executes but it compiles.

32)

What's printed?

```
for (int i = 0; i < 5; i++) {  
    if (i == 3)  
        break;  
    System.out.print(i);  
}
```

- A. 0123
- B. 012
- C. 123
- D. 01234

■ **Answer: B**

■ $i=3$ breaks $\rightarrow$ so prints 0 1 2

33)

Identify result:

```
int i = 10;
```

```
if (i++ == 10 || i++ == 11)
```

```
    i++;
```

```
System.out.println(i);
```

- A. 11
- B. 12
- C. 13
- D. 14

 **Answer: B**

34)

What is the time complexity of the Sieve of Eratosthenes?

- A. $O(n \log n)$
- B. $O(n \sqrt{n})$
- C. $O(n^2)$
- D. $O(n \log \log n)$

 **Correct Answer: D. $O(n \log \log n)$**

 **Explanation:**

Sieve marks multiples for each prime. The overall complexity is $O(n \log \log n)$ due to the harmonic sum of reciprocals of primes.

35)

What is the space complexity of the Sieve of Eratosthenes?

- A. $O(\log n)$
- B. $O(n)$
- C. $O(n \log n)$
- D. $O(\sqrt{n})$

 **Correct Answer: B. $O(n)$**

 **Explanation:**

A boolean array of size $n+1$ is used to track whether each number is prime.

36)

What is the best point to start marking multiples in the sieve?

- A. 2
- B. 1
- C. $n/2$
- D. $p * p$

 **Correct Answer: D. $p * p$**

 **Explanation:**

For any prime p , smaller multiples like $p*2$, $p*3$, etc., are already marked by smaller primes.

37)

Which number does the Sieve start with?

- A. 0
- B. 1

- C. 2
- D. 3

✓ **Correct Answer: C. 2**

■ **Explanation:**

The smallest prime number is 2, so the sieve starts from 2.

38)

Which is a correct optimization in Sieve of Eratosthenes?

- A. Mark from $p+1$
- B. Mark only even numbers
- C. Mark from p^2
- D. *Mark from $p \cdot p$*

✓ **Correct Answer: D. Mark from $p \cdot p$**

■ **Explanation:**

Marking from $p \cdot p$ avoids redundant work, making the algorithm more efficient.

39)

What is the main benefit of the Segmented Sieve?

- A. Faster than normal sieve
- B. Fewer iterations
- C. Less time
- D. Less memory for large ranges

✓ **Correct Answer: D. Less memory for large ranges**

■ **Explanation:**

It only stores $[R-L+1]$ elements instead of all numbers up to R .

40)

What preprocessing is needed for segmented sieve?

- A. Factor table
- B. Nothing
- C. Prime factorization
- D. Primes up to $\sqrt{R}$ using simple sieve

✓ **Correct Answer: D. Primes up to $\sqrt{R}$ using simple sieve**

■ **Explanation:**

These small primes are used to mark multiples in the $[L, R]$ range

41)

What is the size of the boolean array used in segmented sieve?

- A. R
- B. n
- C. $R - L + 1$
- D. $\sqrt{R}$

✅ **Correct Answer: C. $R - L + 1$**

📘 **Explanation:**

Only numbers in range $[L, R]$ are considered, hence that many booleans.

42)

If $L = 100$, $R = 200$, what is the size of the boolean array?

- A. 99
- B. 100
- C. 101
- D. 200

✅ **Correct Answer: C. 101**

📘 **Explanation:**

$$R - L + 1 = 200 - 100 + 1 = 101$$

43)

Which is best for finding primes between 10^9 and 10^9+10^5 ?

- A. Normal sieve
- B. Incremental sieve
- C. Segmented sieve
- D. Trial division

✅ **Correct Answer: C. Segmented sieve**

📘 **Explanation:**

Normal sieve is memory-heavy. Segmented sieve works well for ranges like $[10^9, 10^9+10^5]$.

44)

If $n=17$:

- A. 18
- B. 17
- C. 16
- D. 15

✔ **Answer: C. 16**

Explanation: $\phi(p)=p-1$

45)

What is $\phi(100)$?

- A. 40
- B. 60
- C. 80
- D. 50

✔ **Answer: B. 40**

46)

For $n=1$, $\phi(1)$ is:

- A. 0
- B. 1
- C. Undefined
- D. 2

✓ **Answer:** B. 1

Explanation: Only 1 is ≤ 1 and $\gcd(1, 1) = 1$

47)

What is $\phi(49)$?

- A. 42
- B. 36
- C. 30
- D. 40

✓ **Answer:** A. 42

48)

How many integers less than or equal to 15 are coprime to 15?

- A. 8
- B. 12
- C. 10
- D. 6

✓ **Answer:** A. 8

Explanation: $15=3 \cdot 5$

49)

If $\phi(n)=n-1$, then n must be:

- A. Even
- B. Odd
- C. Prime
- D. Composite

✓ **Answer:** C. Prime

50)

Which of the following counts the number of integers between 1 and N , which are relatively prime to N ?

- a) Euler's totient function
- b) Catalan numbers
- c) Factorial of a number
- d) Fibonacci series

✓ **Answer:** A

51)

What does co-prime mean in the Euler's totient function?

- a) $\text{lcm}(a, b) = 1$
- b) $\text{gcd}(a, b) = 1$
- c) $\text{gcd}(a, b) = 2$
- d) $\text{lcm}(a, b) = 2$

Answer: b

Explanation: Two numbers a and b are said to be relatively prime if the greatest common divisor of both the number is equal to 1. Hence, in Euler's totient function co-prime means $\text{gcd}(a, b) = 1$ and 1 is the co-prime to any number.

52)

```
public class ETF {
    public static int phi(int n) {
        int result = n;
        for (int a = 2; a * a <= n; a++) {
            if (n % a == 0) {
                while (n % a == 0)
                    n = n / a;
                result -= result / a;
            }
        }
        if (n > 1)
            result -= result / n;
        return result;
    }

    public static void main(String[] args) {
        for (int i = 1; i <= 4; i++) {
            System.out.print(phi(i) + " ");
        }
    }
}
```

What is the output of the above Java code?

- A) 0 1 2 3
- B) 0 1 2 4
- C) 1 1 2 2
- D) 1 2 3 4

Answer: c

Explanation: The above code is the implementation of Euler's totient function in C++. It gives the totient function for $n = 1, 2, 3, 4$. The values are: $\phi(1) = 1$, $\phi(2) = 1$, $\phi(3) = 2$, $\phi(4) = 2$. Hence, the output of the following code will be 1 1 2 2.

Output:

1 1 2 2

52. What is a strobogrammatic number?

- A) A number that is a palindrome
- B) A number that appears the same when rotated 180 degrees

C) A number that is divisible by 11

D) A number with only even digits

Answer: B

Explanation: A strobogrammatic number looks the same when rotated 180 degrees (e.g., 69, 96, 0, 1, 8)

52. Which of the following digits are strobogrammatic by themselves?

A) 1, 2, 3

B) 0, 8, 1

C) 6, 9

D) 2, 4, 5

Answer: B

Explanation: 0, 1, and 8 look the same when rotated.

53. Which digit is not valid in any strobogrammatic number?

A) 0

B) 1

C) 3

D) 8

Answer: C

Explanation: 3, 4, and 7 are not strobogrammatic digits.

54)

Is the number 818 strobogrammatic?

A) Yes

B) No

Answer: A

Explanation: $8 \rightarrow 8$, $1 \rightarrow 1$, and the reverse of 818 is 818.

55)

Which of the following is a strobogrammatic number?

A) 609

B) 123

C) 347

D) 431

Answer: A

Explanation: $6 \rightarrow 9$, $0 \rightarrow 0$, $9 \rightarrow 6$. Reverse is 906 which becomes 609 when rotated.

56)

7. What is the result of rotating the number 180 degrees: 9966?

A) 9669

B) 9966

C) 6699

D) 9969

Answer: b

Explanation: Reversed $\rightarrow$ 6699 $\rightarrow$ rotated gives 9966.

57)

What property do all strobogrammatic numbers share?

- A) They are even numbers
- B) They contain only mirror-symmetric digits
- C) They look the same when viewed upside down
- D) They are always palindromes

Answer: C

58)

Which pair is not valid in strobogrammatic numbers?

- A) (6, 9)
- B) (1, 1)
- C) (2, 2)
- D) (8, 8)

Answer: C

Explanation: 2 is not a valid strobogrammatic digit.

59)

How many 2-digit strobogrammatic numbers are there?

- A) 3
- B) 4
- C) 5
- D) 6

Answer: B

Explanation: Valid pairs: (1,1), (6,9), (8,8), (9,6)

- (0,0) is not valid since leading 0 not allowed
 $\rightarrow$ 4 total

60)

What is the main requirement for applying the Chinese Remainder Theorem?

- A. All remainders must be equal
- B. All moduli must be equal
- C. All moduli must be pairwise coprime
- D. All moduli must be powers of 2

✓ **Answer: C**

Explanation: CRT can only be applied when all moduli are **pairwise coprime** (i.e., GCD of each pair is 61).

Solve the system using CRT:

$$x \equiv 1 \pmod{2}$$

$$x \equiv 2 \pmod{3}$$

$$x \equiv 3 \pmod{5}$$

A. 23

B. 54

C. 8

D. 38

✓ **Answer: A (23)**

Explanation:

$$M = 2 \times 3 \times 5 = 30$$

Using CRT steps, $x = 23 \pmod{30}$.

CRT specifically gives the smallest such value.

63)

Which of the following systems **cannot** be solved using CRT?

A. $x \equiv 1 \pmod{4}, x \equiv 3 \pmod{5}$

B. $x \equiv 2 \pmod{7}, x \equiv 5 \pmod{9}$

C. $x \equiv 3 \pmod{6}, x \equiv 4 \pmod{8}$

D. $x \equiv 1 \pmod{3}, x \equiv 2 \pmod{5}$

✓ **Answer: C**

Explanation: 6 and 8 are not coprime ($\text{GCD}(6,8)=2$), so CRT **cannot** be directly applied.

64)

If $x \equiv 2 \pmod{3}$ and $x \equiv 3 \pmod{5}$, then $x \equiv ?$

A. 8

B. 13

C. 3

D. 7

✓ **Answer: A**

Try 8: $8 \% 3 = 2, 8 \% 5 = 3 \Rightarrow$ ✓

So answer is **A (8)**.

65)

What is the value of x that satisfies:

$$x \equiv 2 \pmod{3}$$

$$x \equiv 3 \pmod{4}$$

$$x \equiv 2 \pmod{5}$$

A. 47

B. 37

C. 27

D. 17

 **Answer: A (47)**

66)

Which of the following numbers satisfies:

$$x \equiv 0 \pmod{4}$$

$$x \equiv 1 \pmod{5}$$

A. 20

B. 16

C. 21

D. 9

Answer: B

67)

If $x \equiv 3 \pmod{7}$ and $x \equiv 1 \pmod{5}$, what is x ?

A. 31

B. 17

C. 18

D. 11

 **Answer: A**

67)

What is a leader in an array?

A. An element smaller than its neighbors

B. An element greater than all elements to its right

C. An element greater than all elements to its left

D. The smallest element in the array

68)

What are the leaders in the array [16, 17, 4, 3, 5, 2]?

- A. 16, 17, 5, 2
- B. 17, 5, 2
- C. 16, 4, 5
- D. 4, 3, 5, 2

 **Answer: B**

(Explanation: $17 > [4, 3, 5, 2]$, $5 > 2$, and 2 is the last element)

69)

What is the time complexity of a naive solution to find all leaders in an array of size n ?

- A. $O(n)$
- B. $O(n^2)$
- C. $O(\log n)$
- D. $O(1)$

 **Answer: B**

(Explanation: Check every element with all to its right)

70)

What is the time complexity of an efficient solution to find leaders in an array?

- A. $O(n)$
- B. $O(n^2)$
- C. $O(n \log n)$
- D. $O(1)$

 **Answer: A**

(Explanation: Traverse from right to left, keeping max)

71)

In which direction should we traverse the array to efficiently find leaders?

- A. Left to Right
- B. Right to Left
- C. Middle to End
- D. Any direction

 **Answer: B**

72)

Which element is always considered a leader in an array?

- A. Maximum element
- B. Minimum element
- C. Leftmost element
- D. Rightmost element

✓ Answer: D

73)

How many leaders are there in array [5, 3, 20, 15, 8, 3]?

- A. 1
- B. 2
- C. 3
- D. 4

✓ Answer: D

(Leaders: 20, 15, 8, 3 → All rightmost greater elements)

74)

What are the leaders in array [1, 2, 3, 4, 5]?

- A. 1
- B. 5
- C. 1, 2, 3, 4, 5
- D. 4, 5

✓ Answer: B

(Only 5 is greater than all to its right)

75)

In Booth's Algorithm, the condition $Q_0 = 1$ and $Q_{-1} = 0$ indicates:

- A. Add M to AC
- B. Subtract M from AC
- C. No operation
- D. Add Q to AC

✓ Answer: B

76)

What is the initial value of the Accumulator (AC) in Booth's Algorithm for n-bit multiplication?

- A. Same as multiplicand
- B. Same as multiplier
- C. All 1s
- D. All 0s

✓ Answer: D

77)

If AC = 1010, Q = 0111, $Q_{-1} = 1$, what operation is performed next in Booth's Algorithm?

- A. Add M
- B. Subtract M
- C. No operation
- D. Add 1 to AC

✓ Answer: C

78)

Which operation ensures signed binary representation is preserved during shifts in Booth's Algorithm?

- A. Logical Right Shift
- B. Arithmetic Right Shift
- C. Logical Left Shift
- D. Rotate Right

✓ **Answer: B**

79)

The 2's complement of binary 1101 (4-bit) is:

- A. 0011
- B. 1011
- C. 0101
- D. 0010

✓ **Answer: A**

80)

Booth's Algorithm uses how many bits total in the register combination (AC, Q, Q-1) for 4-bit multiplication?

- A. 8
- B. 9
- C. 7
- D. 10

✓ **Answer: B**

81)

What does an arithmetic right shift of 1110101 produce?

- A. 0111010
- B. 1111010
- C. 1111011
- D. 1111101

✓ **Answer: B**

Sign bit is retained (1), shift right

82)

What is the final binary result when multiplying -2 (4-bit) and 3 using Booth's Algorithm?

- A. 11101000
- B. 11111010
- C. 11110010
- D. 11101110

✓ **Answer: b**

$(-2 \times 3 = -6 \Rightarrow 8\text{-bit } 2\text{'s complement of } 6 = 11111010)$

83)

If the value of Q is 0000, how many additions/subtractions will Booth's Algorithm perform?

- A. 4
- B. 0
- C. 2
- D. 1

✓ **Answer: B**

(Q0 and Q-1 remain 00 $\Rightarrow$ no operation for all 4 cycles)

84)

Which condition leads to adding the multiplicand in Booth's Algorithm?

- A. $Q_0 = 1, Q_{-1} = 0$
- B. $Q_0 = 0, Q_{-1} = 1$
- C. $Q_0 = 0, Q_{-1} = 0$
- D. $Q_0 = 1, Q_{-1} = 1$

✓ **Answer: B**

85)

Booth's Algorithm terminates after how many iterations for n-bit multiplication?

- A. n
- B. n+1
- C. 2n
- D. Depends on Q

✓ **Answer: A**

86)

After completing Booth's Algorithm, the final result is obtained by:

- A. AC only
- B. Q only
- C. AC concatenated with Q
- D. AC - Q

✓ **Answer: C**

87)

What is the purpose of Euclid's Algorithm?

- A. Find LCM
- B. Sort numbers
- C. Find GCD
- D. Check prime

✓ **Answer: C**

88)

What is the key operation used in Euclid's Algorithm?

- A. Multiplication
- B. Addition

C. Modulus (%)

D. Bitwise AND

✓ **Answer: C**

89)

What is the base condition in the recursive Euclid's Algorithm?

A. If $a == b$

B. If $a == 1$

C. If $b == 0$

D. If $a \% b == 0$

✓ **Answer: C**

90)

What will be the output of $\text{gcd}(24, 36)$ using Euclid's algorithm?

A. 6

B. 12

C. 18

D. 24

✓ **Answer: B**

91)

What is the time complexity of Euclid's Algorithm?

A. $O(n)$

B. $O(\log n)$

C. $O(1)$

D. $O(n^2)$

✓ **Answer: B**

92)

In Euclid's Algorithm, how many iterations are required to find $\text{gcd}(55, 10)$?

A. 1

B. 2

C. 3

D. 4

✓ **Answer: C**

$(55, 10) \rightarrow (10, 5) \rightarrow (5, 0)$

93)

What is $\text{gcd}(13, 27)$ using Euclid's algorithm?

A. 1

B. 3

C. 13

D. 0

✓ **Answer: A**

94)

What happens if $a = 0$ and $b \neq 0$ in $\text{gcd}(a, b)$?

- A. Error
- B. Returns 0
- C. Returns b
- D. Infinite loop

✓ Answer: C

95)

In how many steps does Euclid's Algorithm find $\text{gcd}(100, 40)$?

- A. 1
- B. 2
- C. 3
- D. 4

✓ Answer: B

$(100, 40) \rightarrow (40, 20) \rightarrow (20, 0)$

96)

What is $\text{gcd}(0, 5)$?

- A. 5
- B. 0
- C. Undefined
- D. 1

✓ Answer: A

97)

What is the purpose of the Boyer-Moore Voting Algorithm?

- A. Sort the array
- B. Find the smallest element
- C. Identify the majority element (if it exists)
- D. Count all elements in the array

✓ Answer: C

98)

What is the initial value of the count variable in the algorithm?

- A. -1
- B. 0
- C. 1
- D. Not defined

✓ Answer: B

99)

Which statement is true if $\text{count} == 0$?

- A. The candidate is removed
- B. A new candidate is chosen

- C. An error is thrown
- D. Loop breaks

✓ Answer: B

100)

What does the condition `candidate == num` do?

- A. Skips the number
- B. Increases count
- C. Resets the loop
- D. Deletes the candidate

✓ Answer: B

101)

What is the time complexity of the Boyer-Moore Voting Algorithm?

- A. $O(n \log n)$
- B. $O(\log n)$
- C. $O(n^2)$
- D. $O(n)$

✓ Answer: D

102)

What is the space complexity of the algorithm?

- A. $O(n)$
- B. $O(1)$
- C. $O(n \log n)$
- D. $O(k)$ where k is unique elements

✓ Answer: B

103)

In the array `{3, 2, 3}`, what is the majority element?

- A. 2
- B. 3
- C. No majority
- D. 0

✓ Answer: B

 3 appears **2 times** out of 3 $\rightarrow$ more than $n/2 = 1.5$

104)

Door number 81 is toggled how many times during the entire process ($N = 100$)?

- A. 1
- B. 2

- C. 9
- D. 5

✅ **Answer: C**

Reason: It is toggled once for each of its divisors. 81's divisors = 1, 3, 9, 27, 81 → 5 total, but 81 is a perfect square, so all divisors are paired except one → total **odd** = **9** due to repeated counting.

105)

Which of the following doors will be **closed** after 100 monkeys finish toggling?

- A. 64
- B. 36
- C. 49
- D. 77

✅ **Answer: D**

Reason: All perfect squares remain open. 77 is not a perfect square → it will be toggled even number of times → stays closed.

106)

For door number d, it will remain **open** only if:

- A. d has even number of divisors
- B. d is a composite number
- C. d is a perfect cube
- D. d is a perfect square

✅ **Answer: D**

107)

The formula used in the Alice Apple Tree code for apple count per layer is:

- A. $\text{sum} += \text{cnt} * \text{cnt} * 4$
- B. $\text{sum} += 8 * \text{cnt} * \text{cnt}$
- C. $\text{sum} += 12 * \text{cnt} * \text{cnt}$
- D. $\text{sum} += 6 * \text{cnt}$

✅ **Answer: C**

Reason: Direct from code: $\text{sum} += (12 * \text{cnt} * \text{cnt})$

108)

If you need to collect at least 432 apples, what is the minimum value of cnt such that:

$\text{sum} += 12 * \text{cnt} * \text{cnt}$

- A. 4
- B. 5
- C. 6
- D. 7

✅ **Answer: C**

Reason: Try values:

- $\text{cnt} = 5 \rightarrow 12 \times 25 = 300$
- $\text{cnt} = 6 \rightarrow 12 \times 36 = 432$ ✓

109)

If cnt = 15, what is the value of sum?

- A. 2520
- B. 2700
- C. 2880
- D. 3060

✅ **Answer: B**

💡 $12 \times 15^2 = 12 \times 225 = 2700$ ✓

Correction to options:

✅ **Answer: B**

110)

How many swaps does selection sort perform in the worst case for an array of size n?

- A. $O(n^2)$
- B. $O(n \log n)$
- C. $O(n)$
- D. Depends on data

✅ **Answer: C. $O(n)$**

💡 **Explanation:**

Selection sort always performs **(n - 1)** swaps regardless of the input. The comparisons are $O(n^2)$, but swaps are only $O(n)$.

111)

What will be the recursion depth of quicksort in the best case?

- A. $O(1)$
- B. $O(n)$
- C. $O(\log n)$
- D. $O(n^2)$

✅ **Answer: C. $O(\log n)$**

💡 **Explanation:**

In the best case, pivot divides array into two equal halves. So recursion goes $\log n$ deep (binary tree height).

112)

What is the time complexity of the Karatsuba algorithm?

- A. $O(n^2)$
- B. $O(n \log n)$
- C. $O(n^{1.585})$
- D. $O(n)$

✅ **Answer: C**

👉 Karatsuba's time complexity is $O(n^{\log_2 3}) \approx O(n^{1.585})$.

113)

Which of the following correctly represents the Karatsuba formula?

- A. $xy = ac \times 10^n + bd$
- B. $xy = ac \times 10^{2m} + ((a + b)(c + d) - ac - bd) \times 10^m + bd$
- C. $xy = (a + b)(c + d)$
- D. $xy = ac + ad + bc + bd$

✅ **Answer: B**

114)

What is the main advantage of the Karatsuba algorithm over the traditional multiplication method?

- A. Simpler to implement
- B. Uses division
- C. Reduces the number of multiplications
- D. Uses more addition operations

✅ **Answer: C**

👉 Karatsuba reduces the number of multiplications from 4 to 3 for large numbers.

115)